CANTRIP Dataset Documentation for Is This a Time for Irony? Detection Ironic-Response Affordance in Dialogues

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Abstract

This repository document contains the complete CANTRIP Data Statement, representative instances for both proxy classes, and the full post-editing procedure used to validate and normalize the generated dialogues.

I. CANTRIP Data Statement

We provide a data statement of CANTRIP, as proposed by [1]. We first provide sample instances for both response-derived proxy classes (0, 1); the construction pipeline is summarized in the main paper.

CANTRIP$_{\text{CSC}}$ *negative proxy* instance

Dialogue

$\mathbf{M_0^{U_1}}$: Just got back from the market and stocked up on some fresh bread.
$\mathbf{M_0^{U_2}}$: Ah, nice! What was the deal with the baker today?

$\mathbf{M_1^{U_1}}$: Not much. They had a few new varieties, but I went with the usual.
$\mathbf{M_1^{U_2}}$: Same here. There's nothing like a fresh loaf straight from the oven.

$\mathbf{M_2^{U_1}}$: It will all be stale in a week anyway, but the shops will still be open...

Response-derived Affordance Proxies on $\mathbf{M_0^{U_1} + M_0^{U_2} + \ldots + M_2^{U_1}}$

- Majority Vote: 0
- Male: 0 Female: 0
- GenX: 0 GenY: 0 GenZ: 0

CANTRIP$_{\text{CSC}}$ *positive proxy* instance

Dialogue

$\mathbf{M_0^{U_1}}$: I just saw an announcement about the weekend.
$\mathbf{M_0^{U_2}}$: Oh, yeah? What about it?

$\mathbf{M_1^{U_1}}$: It looks like the weather is going to be great!
$\mathbf{M_1^{U_2}}$: That sounds awesome, perfect for a beach day!

$\mathbf{M_2^{U_1}}$: That's fabulous Friday news!

Response-derived Affordance Proxies on $\mathbf{M_0^{U_1} + M_0^{U_2} + \ldots + M_2^{U_1}}$

- Majority Vote: 1
- Male: 1 Female: 0
- Boomer: 1 GenZ: 1 GenY: 0

a) Curation Rationale.: CANTRIP is constructed through systematic augmentation of two existing irony and sarcasm detection datasets: MultiPICo (MP) and the Conversational Sarcasm Corpus (CSC). We extend single-turn or minimally contextualized conversational exchanges into synthetic multi-turn dialogues to enable the investigation of contextual cues associated with ironic-response affordance proxies. MP consists of post-reply pairs extracted from Twitter and Reddit, annotated by multiple observers following a perspectivist framework. For details on MP's original collection methodology, we refer readers

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to [2]. CSC comprises hand-written conversational contexts paired with naturally occurring responses, labeled by both authors and third-party annotators. For comprehensive information about CSC's data collection process, see [3]. Our augmentation process employs Llama3.1-8B to generate preceding conversational turns, creating three-turn dialogues. Generated content was evaluated by humans using a taxonomy adapted from CheckEval [4], followed by automatic quality filtering using Gemma2-9B as an LLM-as-judge. Only instances scoring at least 8 out of 16 points across four dimensions (Coherence, Engagingness, Groundedness, and Naturalness) were retained in the final dataset.

b) Language Variety.: The dataset comprises predominantly US English from social media platforms (Twitter and Reddit) inherited from MP, and hand-written conversational English from the CSC. The augmented multi-turn dialogues generated by Llama3.1-8B reflect plausible conversational English patterns.

c) Author and Annotator Demographics.: `CANTRIP` inherits annotator demographics from its source datasets, MP [2] and CSC [3]. We refer to the original papers for complete demographic information. The human evaluation of `CANTRIP`'s generated dialogues was conducted by four of this paper's authors. The automatic evaluation using Gemma2-9B as a judge does not involve human demographic factors.

d) Text Characteristics.: `CANTRIP` consists of three-turn dialogues where the final utterance stems from MP (social media post-reply pairs from Twitter and Reddit) or CSC (descriptive extra-linguistic contexts paired with responses). The preceding two turns are synthetically generated by Llama3.1-8B to produce coherent conversational contexts. Labels indicate whether the context affords an ironic response , derived from the original irony/sarcasm annotations via the mapping function detailed in the Methodology section of the main paper. `CANTRIP` inherits the perspectivist annotations from its source datasets, including aggregated majority vote labels, gender-specific annotations, generational annotations, and relationship annotations where available.

e) Provenance.: We refer to the original MP[1] [2] and CSC[2] [3] papers for additional documentation.

II. `CANTRIP` Post-Editing Details

In Figure 1, we show an example of dialogue augmentation for both CSC and MP. To ensure consistency and quality of the generated dialogues, we applied a dedicated post-processing pipeline to the raw outputs produced by the LLMs. Post-processing focused on parsing, validation, and normalization of the generated JSON. Given that LLM outputs may contain malformed or partially invalid JSON, we applied a series of cleaning steps, including the removal of illegal trailing commas, normalization of whitespace and newline characters, and correction of improperly escaped strings. We then validated the parsed content by enforcing a fixed dialogue structure composed of alternating speaker turns (U_1, U_2), discarding outputs with missing, duplicated, or repeated utterances. Dialogues with an incorrect number of turns or repeated utterances after normalization were filtered out. All valid dialogues were then converted into a canonical representation, where each turn is encoded as a single U_1–U_2 pair. In addition, for the CSC variant, we removed speaker prefixes (e.g., strings starting with `"Steve:"`), which the model occasionally introduced despite not being part of the target format. Finally, we discarded dialogues with insufficient contextual content. In particular, for the MultiPICo subset, we removed instances where either the post or the reply consisted of a single word or only punctuation, while for CSC we applied the same filtering criterion to the reply alone. This final filtering step ensured that all retained dialogues contained a minimal amount of meaningful linguistic context.

References

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[4] Y. Lee, J. Kim, J. Kim, H. Cho, J. Kang, P. Kang, and N. Kim, "CheckEval: A reliable LLM-as-a-judge framework for evaluating text generation using checklists," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, eds.), (Suzhou, China), pp. 15782–15809, Association for Computational Linguistics, Nov. 2025.

[1] https://huggingface.co/datasets/Multilingual-Perspectivist-NLU/MultiPICo

[2] https://github.com/CoPsyN/CSC

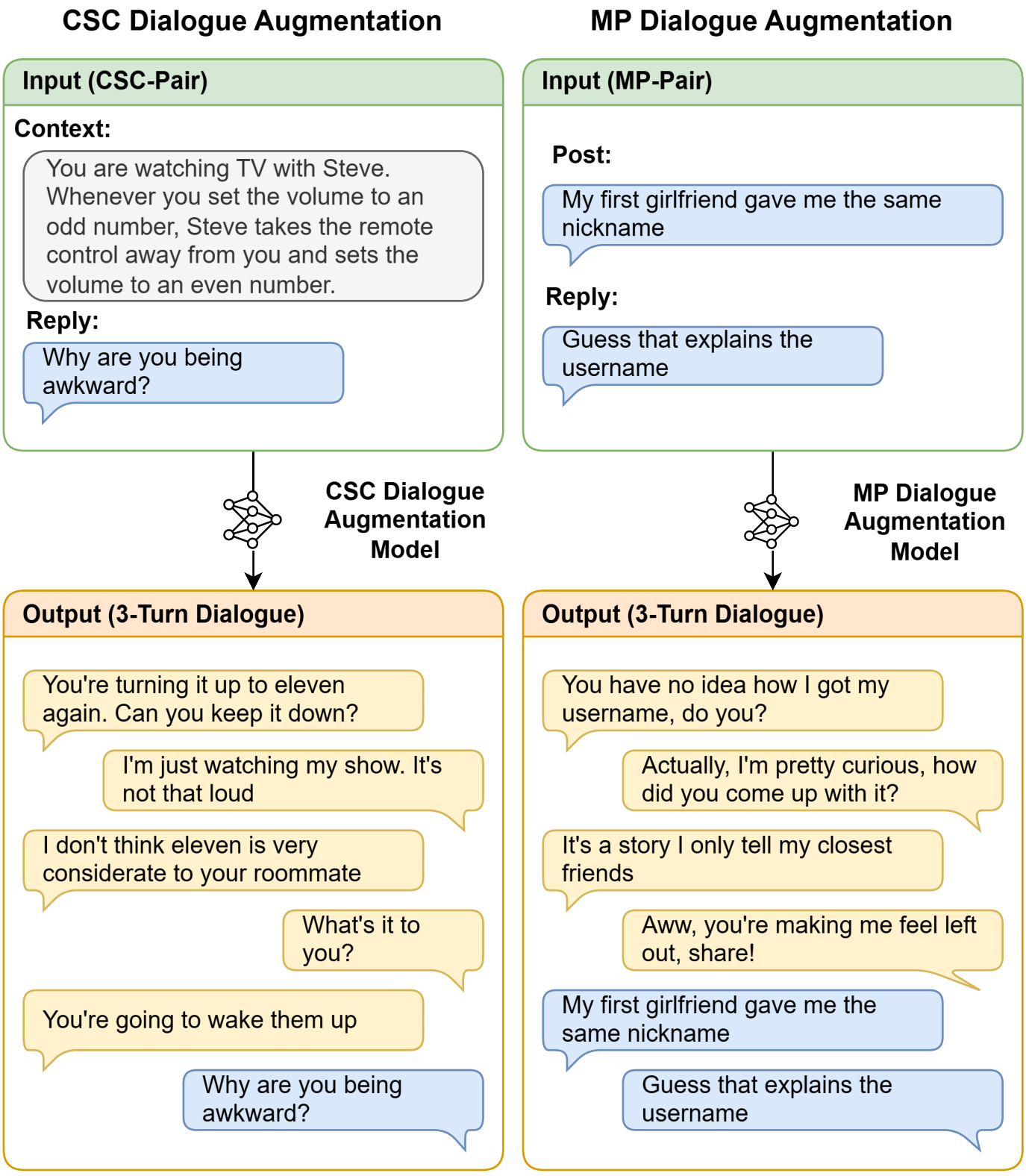


Fig. 1. Example of dialogue augmentation for MP and CSC. For MP, the input consists of a post-reply pair, while for CSC the input is a dialogue context along with a user response. The resulting augmented dialogue follows the same structure in both datasets.